Koushik Maity

Roll No.: 252123014
M.Sc.- Mathematics and Computing
Indian Institute Of Technology, Guwahati

+91-8926875192
koushikmaity576@gmail.com
k.maity@iitg.ac.in
Github | LinkedIn

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Sc. Mathematics and Computing	Indian Institute of Technology, Guwahati	8.30 (Current)	2025-Present
B.Sc. (Hons.) Mathematics	Vidyasagar University	7.82	2021-2024
Senior Secondary	WBCHSE Board	81.60%	2021
Secondary	WBBSE Board	89%	2019

Projects

- **Banking Transaction & Account Management System** *August 2026 – Present*
  Prof. Kalpesh Kapoor, Department of Mathematics, IIT Guwahati Github
  – Designed and implemented a normalized **PostgreSQL database (11 tables)** for a role-based banking system supporting customers, employees, and admins, enforcing data integrity via **keys, constraints, and a unified transaction model** with full audit tracking.
  – Built **ACID-compliant fund transfer and loan payment workflows** using **triggers, stored procedures, and multi-step transactions with rollback**, designed for integration with a **Flask-based frontend**.
- **Customer Loan Conversion Prediction** *June 2026 – July 2026*
  Self-Directed Project Website | Github
  – Built an **end-to-end classification system** on **5,000 customer records** with a **9.6% positive class**, comparing **7 ML algorithms** using **stratified 5-fold cross-validation** and **F1-score**.
  – Applied **permutation-based feature selection** and **Optuna hyperparameter tuning**; optimized **Gradient Boosting** using **Income, Education, Family, and CCAvg**, improving mean CV F1 from **92.74% to 93.93%**.
  – Serialized the complete **Scikit-learn pipeline** using **Joblib**, developed an interactive **Streamlit application**, and deployed it publicly on **Render** for real-time loan-conversion prediction.
- **Gurgaon Real Estate Analytics, Price Prediction & Recommendation** *May 2026 – July 2026*
  Self-Directed Project Github
  – Built an **end-to-end real estate analytics system** using **3,500+ property listings** across approximately **100 Gurgaon sectors**, integrating property, pricing, and location-based information.
  – Engineered features from **area, floor, furnishing, luxury, and location** attributes; handled **missing values, duplicates, and outliers** and evaluated multiple **regression models** using cross-validation with $\mathbf{R^2}$, **MAE**.
  – Developed an interactive **Streamlit application** combining **price prediction, Plotly geographic analytics, and similar-property recommendations**.
- **Unsupervised Health Data Analysis** *April 2026 – May 2026*
  Self-Directed Project Github
  – Processed **NHANES health data** through **multi-table merging**, **missing-value imputation**, noise-code handling, feature engineering, and scaling; applied **PCA**, **K-Means**, and **DBSCAN** for dimensionality reduction and population subgroup analysis.
  – Segmented the population into **10 clusters** using **K-Means** and profiled demographic and socioeconomic characteristics to identify **interpretable population-level insights and outlier profiles**.

Technical Skills

- **Programming**: C, C++, Python
- **Libraries/Frameworks**: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
- **Mathematical Software**: MATLAB*
- **Database/Query Languages**: SQL, MySQL, PostgreSQL*
- **Tools**: Power BI*, Google Colab, Jupyter Notebook, VS Code, Streamlit
- **Operating Systems**: Windows, Linux*

** Elementary proficiency*

Key courses taken

- **Mathematics**: Linear Algebra, Discrete Mathematics, Differential Equations, Optimization Techniques, Real Analysis, Numerical Analysis, Mathematical Modelling
- **Statistics**: Probability and Statistics
- **Computer Science**: Computer Programming, Data Structures and Algorithms, Machine Learning* (Swayam-NPTEL), Theory of Computation*, Database Management System*

**To be completed by Nov 2026*

Achievements

- **CUET(PG) 2025**, Qualified in Mathematics *2025*
- **1st Position in Mathematics**, B.Sc. (Hons.), Mathematics Department, Kharagpur College *2024*
- **Swami Vivekananda Merit-cum-Means Scholarship**, Government of West Bengal *2022–2024*

Extracurricular Activities

- **Peer Mentor (Saathi)**, Counselling Cell, IIT Guwahati *[Aug. 2026 – Present]*
  – Mentored **first-year students** through academic guidance, campus orientation, and support while addressing concerns and connecting them with relevant **institute resources**.